SiteWise AI: Deep Multi-Stage Hierarchical Land Cover Classification and Spatial Suitability Screening for Earth Observation Imagery

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Abstract— Automated satellite Earth observation (EO) analysis is critical for municipal zoning, agricultural resource allocation, environmental conservation, and civil infrastructure planning. However, existing methodologies encounter significant operational challenges: standard local convolutional neural networks (CNNs) struggle with disjoint spatial structures (e.g., divided water networks, fragmented agricultural parcels, or road cuts), pixel-wise segmentations are computationally burdensome for real-time edge screening, and raw land-use classification outputs do not directly translate into actionable development decisions. In this paper, we present SiteWise AI—an integrated, multi-stage hierarchical deep learning framework for satellite land-cover intelligence and multi-criteria site suitability screening. SiteWise AI incorporates a multi-model primary classification stage comprising EfficientNetB0, U-Net, and a novel Pattern-Affinitive Propagation Network (PAPNet). PAPNet utilizes non-local Pattern Affinity Modules (PAM) with learnable residual gating to capture long-range semantic correlations across fragmented terrains. In parallel, an 8-dimensional multispectral and optical feature extractor computes surface telemetry (vegetation index, water absorption, built-up density, linear contrast, scene luminance). These composite representations feed an 8-head multi-task neural recommendation engine (Model 3) that scores parcels across 8 distinct real-world development categories. Furthermore, to enable deployment on memory-constrained cloud environments (such as Render.com 512MB RAM free tier), we demonstrate live graph optimization into ONNX and TFLite formats, reducing runtime memory footprint by 73% (from 450MB to ~120MB) and achieving sub-30ms inference on standard CPUs without loss of accuracy.

Index Terms— Remote Sensing, Sentinel-2, EuroSAT, DeepGlobe, Pattern Affinity Modules, EfficientNet, U-Net, Site Suitability, Edge Deployment, Model Quantization, ONNX Runtime.

I. INTRODUCTION

Recent advancements in multi-spectral satellite constellations—most notably the European Space Agency's Sentinel-2 mission—have democratized access to high-revisit Earth observation data. Despite petabytes of freely available remote sensing imagery, bridging the semantic gap between raw optical pixels and practical spatial development decisions remains an open challenge [1]. Conventional land-cover analysis workflows typically decouple terrain classification from downstream civil engineering assessments, requiring manual GIS post-processing and subjective domain heuristics [2].

Moreover, convolutional neural networks (CNNs) applied to high-resolution satellite tiles suffer from fixed receptive fields. In satellite imagery, semantic entities such as agricultural patches, forest reserves, river channels, and transportation corridors often exhibit disjoint spatial coherence [3]. Standard convolutions require excessively deep architectures to correlate spatially separated parcels, inducing severe computational overhead and spatial blurring [4].

To resolve these limitations, this paper introduces SiteWise AI, an end-to-end multi-stage deep learning platform featuring: (1) multi-model flexibility allowing dynamic switching between compound-scaled EfficientNetB0, pixel-wise semantic U-Net, and attention-guided PAPNet; (2) automated optical surface telemetry extraction; (3) multi-criteria neural development suitability screening; and (4) ultra-low memory optimization for deployment on restricted cloud runtimes.

II. RELATED WORK

Remote sensing classification has progressed from classical spectral index thresholding (e.g., NDVI, NDWI) and support vector machines [5] to modern deep transfer learning. Helber et al. established the EuroSAT benchmark [6], evaluating deep

residual baselines across 10 land cover classes. Tan and Le introduced EfficientNet [7], proving that compound scaling of depth, width, and resolution yields optimal accuracy with minimal parameter counts.

For spatial segmentation, Ronneberger et al. pioneered U-Net [8], whose symmetrical encoder-decoder with skip connections has become the standard for remote sensing pixel mapping [9]. In parallel, non-local neural networks and self-attention mechanisms [10], [11] have emerged to model long-range contextual relationships. However, prior studies rarely integrate these models into an end-to-end decision system or address cloud-edge RAM constraints under 512MB limits.

III. PROPOSED ARCHITECTURE & METHODOLOGY

A. Pipeline Overview

The SiteWise AI inference pipeline operates sequentially: (1) Satellite Tile Input (64x64x3 RGB); (2) Stage 1 Primary Classification via chosen architecture (EfficientNetB0, PAPNet, or U-Net); (3) Stage 2 Surface Telemetry (8 optical indices); (4) Feature concatenation into an 18-element normalized vector; (5) Stage 3 Multi-Objective Recommendation via Model 3, generating calibrated scores across 8 development sectors.

B. Pattern-Affinitive Propagation Network (PAPNet)

PAPNet addresses disjoint spatial parcel coherence through dedicated Pattern Affinity Modules (PAM). Given intermediate feature map X in R^{H x W x C}, PAM projects linear query, key, and value representations: Q = Conv_q(X), K = Conv_k(X), V = Conv_v(X). The non-local spatial affinity matrix A is defined as:

A_{i,j} = exp((Q_i · K_j^T) / sqrt(d_k)) / sum_m exp((Q_i · K_m^T) / sqrt(d_k))

The affinity-propagated feature P = A · V is projected via Conv_proj(P) and blended with the identity input using a learnable residual gating parameter gamma: Y = X + gamma * Conv_proj(P). Initializing gamma at zero ensures smooth gradient flow during initial training epochs.

C. EfficientNetB0 and U-Net Architectures

EfficientNetB0 leverages Mobile Inverted Bottlenecks (MBConv) with Squeeze-and-Excitation (SE) optimization, fine-tuned in a three-phase schedule (frozen backbone -> top layers -> end-to-end cosine decay) to achieve stable convergence over EuroSAT [7]. For fine-grained pixel-level spatial queries, U-Net employs an encoder-decoder topology with skip connections trained on DeepGlobe imagery, generating pixel-level terrain masks and surface ratios [8].

D. Multi-Criteria Recommender (Model 3)

Model 3 accepts the 18-element composite vector (10 class probabilities + 8 visual surface features), scaled via standard normal deviation parameters. An 8-head multi-task neural network with sigmoid activation evaluates multi-objective utility across Residential, Commercial, Industrial, Educational, Healthcare, Agricultural, Transportation, and Green Conservation sectors simultaneously.

IV. EXPERIMENTAL SETUP & RESULTS

Experiments were conducted on the EuroSAT dataset (27,000 geo-referenced Sentinel-2 tiles across 34 European nations, 10 classes) and the DeepGlobe dataset (1,146 land-cover tiles, 7 pixel classes). Data was partitioned into 70% training, 15% validation, and 15% held-out test splits.

Architecture	Parameters	Test Acc.	Macro F1	CPU Latency	Memory (MB)
EfficientNetB0 (Keras)	5.3M	94.8%	0.946	42 ms	450 MB
PAPNet (ONNX)	7.2M	95.4%	0.952	26 ms	118 MB
U-Net (DeepGlobe)	31.0M	89.2% (mIoU 0.718)	0.784 (Dice)	68 ms	145 MB

As demonstrated in Table I, PAPNet delivers the highest classification accuracy (95.4%) while outperforming standard CNNs by +3.8% on topologically fragmented categories such as waterways and road corridors.

V. EDGE DEPLOYMENT & CLOUD OPTIMIZATION

Deploying deep neural architectures to free-tier cloud containers (e.g., Render.com with a strict 512MB RAM ceiling) commonly triggers out-of-memory SIGKILL termination due to heavy TensorFlow runtime overhead (>350MB). To resolve this, all trained models were serialized live into Open Neural Network Exchange (ONNX) and TensorFlow Lite (TFLite) formats.

Utilizing the ONNX Runtime with CPUExecutionProvider and single-threaded inter-op pooling reduced container startup RSS memory from 450MB to ~120MB, leaving over 380MB of safety buffer and reducing latency by 38%.

VI. CONCLUSION & FUTURE SCOPE

SiteWise AI demonstrates a comprehensive, mathematically grounded framework uniting satellite image classification, non-local pattern propagation, spatial telemetry, and automated zoning recommendation. By optimizing model runtimes for low-memory cloud hosting, the platform provides scalable, accessible Earth observation intelligence. Future work will investigate multi-temporal change detection and multispectral infrared band integration.

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